Khwaja Yunus Ali University

School of Science and Engineering

Department of Computer Science and Engineering

Final Examination: Fall Semester 2018

Program: B.Sc. in CSE (Batch: 7^{th}) 1^{st} Year 1^{st} Semester

Course Title: Physics – I

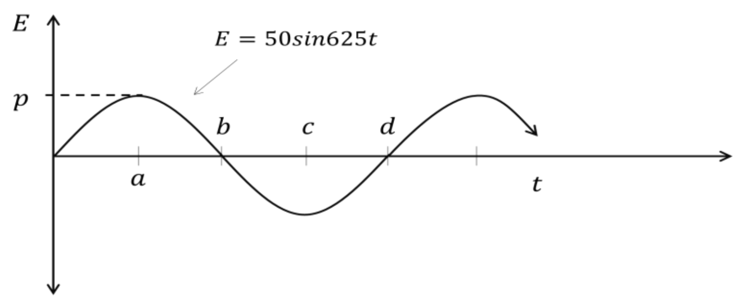
Course Code: PHY 1101

Total Time: 2 hours Full Marks: 40

Group-B

[Answer any 5 (Five) questions including No. 1.] **5 × 6 = 30**

1. See the AC signal carefully and answer the questions that follows –



a) Define Form factor of AC. 1

b) Calculate the r.m.s. value, average value and frequency of above signal. 4

c) Indicate the values of points a, b, c, d & p in the diagram. 1

2. a) Why the AC 220volt is more dangerous than DC 220volt. 2

b) Magnetic flux in a coil of 400 turns changes from $8 \times 10^{-5} wb$ to $50 \times 10^{-5} wb$ in $3 \times 10^{-3} sec$. Calculate the induced emf in the coil. 4

3. a) State the Faraday's 1^{st} law of electromagnetic induction. 1

b) How is Lenz's Law related to the conservation of energy? 2

c) If the current in the primary coil is changed from $6A$ to $1A$ in $0.05 sec$, then emf $5V$ is induced in secondary coil. What is the mutual inductance of the coil? 3

4. a) Write a short note on hall effect. 2

b) Explain the forces between two parallel current carrying conductors. 4

5. a) How the torque produced on a current carrying coil in magnetic field. 3

b) A coil of 20 turns has an area of $800 mm^2$ and bears a current of 0.5A. It is placed with its plane parallel to magnetic field of intensity 0.3T. Determine the torque and magnetic moment on the coil. 3

6. a) Define Lorenz force. 1

b) Write down the conclusion of the Oersted's Experiment. 2

c) A straight horizontal segment of copper wire carries $i = 28A$. What is the magnitude and direction of the magnetic field needed of float the wire, that is, to balance its weight? Its linear mass density is $46.6 g/m$. 3

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Group-A

Time: 25 min | **[Answer all questions]** | **0.5 × 20 = 10**

Student Name:	**ID No.:**

1. What is the value of magnetic permeability of free space?
   a) $8.85 \times 10^{-12} C^2 N^{-1} m^{-2}$ b) $9\times 10^{9} Nm^2C^{-2}$
   c) $4\pi \times 10^{-7} TmA^{-1}$ d) $1.6 \times 10^{-19} C$
   e) $9.1 \times 10^{-31} kg$

2. Magnitude of the magnetic field is ………………….. to the strength of current.
   a) proportional b) equal
   c) inversely proportional d) independent
   e) less than

3. The outward direction of magnetic field in a plane paper is indicated by -
   a) alpha sign b) beta sign
   c) tick sign d) cross sign
   e) dot sign

4. According to the Ampere's law, $\oint_l \bar{B}.\bar{dl} = ?$
   a) i b) $\frac{i}{\mu_0}$
   c) $\mu_0 i$ d) $\frac{\mu_0}{i}$
   e) μ_0

5. If the angle between a current carrying wire and magnetic field is zero, then the value of magnetic force will be -
   a) 0 b) 1
   c) 2 d) 4
   e) 8

6. The relation between magnetic field and magnetic flux is –
   a) $B = \frac{\varphi}{2} A$ b) $B = \frac{i}{\varphi} A$
   c) $B = \frac{A}{\varphi}$ d) $B = \frac{\varphi}{A}$
   e) $B = A\varphi$

7. What is the unit of the magnetic flux?
   a) T
   b) wb
   c) $Gauss$
   d) $Volt$
   e) A

8. Number of flux that changes per second in a coil for the production of 1volt potential difference in that coil is called …………………
   a) 1 weber
   b) 1 Tesla
   c) 1 Gauss
   d) 1 Ampere
   e) 1 Henry

9. $1\,Tesla = ?$
   a) $10\ \ Gauss$
   b) $10^2 Gauss$
   c) $10^4 Gauss$
   d) $10^6 Gauss$
   e) $10^{10} Gauss$

10. What is the unit of magnetic moment?
    a) μF
    b) TmA
    c) Tm
    d) Am
    e) Am^2

11. The magnetic field of a current carrying wire in the inside of itself is -
    a) $10^{-7} TmA^{-1}$
    b) $4\pi\ TmA^{-1}$
    c) $4\pi \times 10^{-7} TmA^{-1}$
    d) π
    e) zero

12. Two parallel wire carrying current in opposite direction, …………..
    a) independent to each other.
    b) join each other.
    c) attract each other.
    d) repel each other.
    e) burn each other.

13. Current carrying solenoid behave like a -
    a) toroid
    b) magnet
    c) insulator
    d) semiconductor
    e) source

14. The number of charges per unit volume of a conductor can be determined by -
    a) Biot-savart’s law
    b) Ampere’s law
    c) Hall effect
    d) Lorenz force
    e) Lenz’s law

15. The frequency of an alternating current is 50hz, how long will take to reach the peak from the zero?
    a) $0.005\ sec$
    b) $0.500\ sec$
    c) $0.25\ sec$
    d) $0.15\ sec$
    e) $1.5\ sec$

16. What is the value of form factor of an alternating current?

a) 0.5 b) 1.11
c) 2.11 d) 3.11
e) 4.11

17. What is the unit of the e.m.f.?

a) T b) wb
c) $Gauss$ d) $Volt$
e) A

18. The value of proportionality constant of Newman's law is –

a) 1 b) 4π
c) π d) 10^{-10}
e) 10^{-7}

19. Which one is right for the relationship between flux and inductance?

a) $\varphi = NLi$ b) $\varphi = NMi$
c) $\varphi = Ni$ d) $\varphi = NL$
e) $N\varphi = Li$

20. The average value for the full cycle of an alternating current is -

a) $0.637I_0$ b) 0.637
c) $Zero$ d) $0.707I_0$
e) 0.707

Invigilator Signature

Examiner Signature